NegLaB Research Colloquium – October 13, 2026

Conjunction, negation, and how focus affects their scope

When De Morgan is not the answer

Andrea Matticchio

UMass Amherst & Goethe University Frankfurt

1 | Relevance

2 Negated conjunction

3 Focus and alternatives

4 Modal intervention

5 Conclusions

Question-answer congruence

A puzzle for semantics: Which sentences answer which questions?

- Classical answer: **relevance**; *Lewis 1988; Groenendijk & Stokhof 1984; Roberts 2012*
- Still the main factor in recent work. *Feinmann 2026; Benbaji-Elhadad & Doron 2026*

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Today's thesis

Relevance is not what natural language uses to decide congruence.

Questions partition the logical space

Two assumptions:

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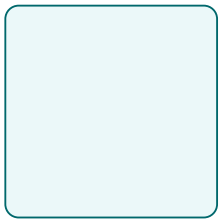
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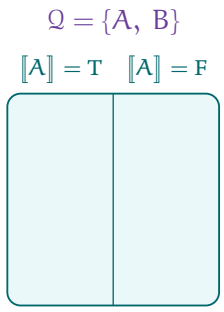
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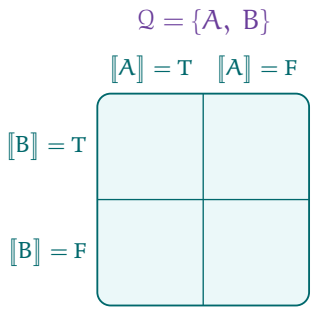
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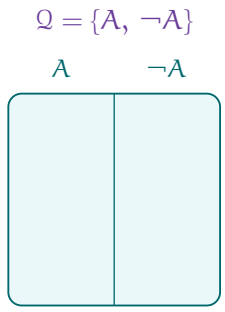
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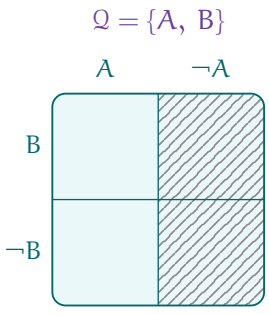
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Relevance

One of many definitions

P is relevant to Q iff P rules out at least one whole cell of Q. *Groenendijk & Stokhof 1984*



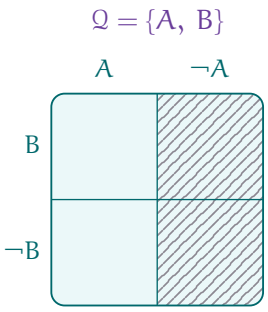
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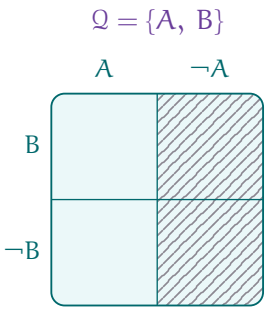
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2. No need to leave exactly one cell.
   a or b answers *a, b, or c*?



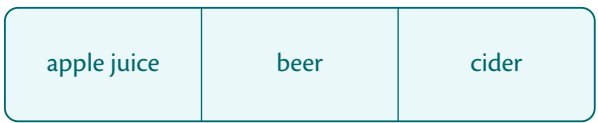
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No need to pick just one cell

CONTEXT: *Apple juice, beer, cider. Taro had exactly one drink.*

(1) Q: Which drink did Taro have? $\mathbb{Q} = \{A, B, C\}$

A: He had the apple juice or the beer. + implicatures $(A \vee B) \wedge \neg C$

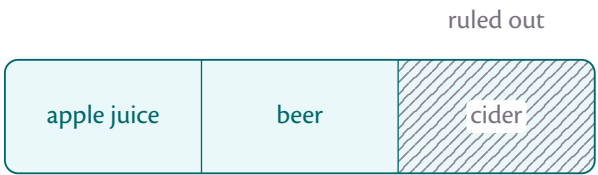


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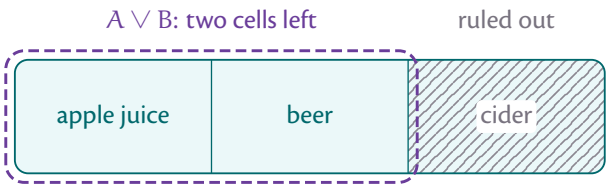


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relevant without resolving the question

1 Relevance

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3 Focus and alternatives

4 Modal intervention

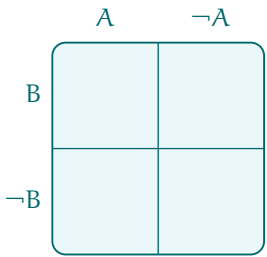
5 Conclusions

De Morgan

A classical equivalence: negated conjunction has the same truth conditions as a disjunction.

(2) Taro didn't have both the apple juice and the beer. $\neg(A \wedge B)$

(3) Either Taro didn't have the apple juice, or he didn't have the beer. $\neg A \vee \neg B$



using *both* to remove the homogeneous reading of conjunction

De Morgan

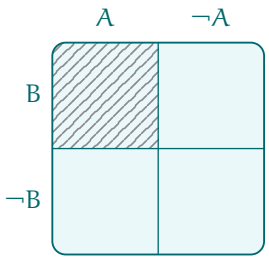
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$$\neg(A \wedge B) \iff \neg A \vee \neg B$$

using *both* to remove the homogeneous reading of conjunction



same cell ruled out

Different answers despite being equivalent

(4) Q: Which drinks did Taro not have?

A: Either he didn't have [the apple juice]$_F$, or he didn't have [the beer]$_F$. $\neg A \vee \neg B$

A′: He didn't have both [the apple juice]$_F$ and [the beer]$_F$.

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Intuition

A′ only has a reading where conjunction scopes above negation.

The experiment

- Acceptability judgment task in Italian, 1–7 Likert scale;
- 2×2 design, within participants and within items;
- 38 participants, 36 items.

joint work with Maik Thalmann (Ruhr University Bochum)

	polarQ	whQ
nand	Did he do both A and B, or not?	Which ones did he not do?
or	Did he do A or B, or not?	Which ones did he do?

with ignorance continuations, which remove the $\neg A \wedge \neg B$ reading

Items

(5) a. Did Peter hide both the necklace and the bracelet, or did he not? polarQ-nand
No, he didn't hide both the necklace and the bracelet, but I don't remember which he didn't hide.

b. Which pieces of jewelry did Peter not hide? whQ-nand
He didn't hide both the necklace and the bracelet, but I don't remember which he didn't hide.

c. Did Peter hide the necklace or the bracelet, or did he not? polarQ-or
Yes, he did hide the necklace or the bracelet, but I don't remember which he hid.

d. Which pieces of jewelry did Peter hide? whQ-or
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An example trial

Quanto è accettabile la risposta alla domanda?

C'erano tre gioielli sul comodino: una collana, un braccialetto e un anello.
Pietro, che aveva intenzione di nascondere alcuni dei gioielli, ma non di nasconderli tutti, riceverà una ricompensa se ha nascosto sia la collana che il braccialetto.



Ma quindi Pietro ha nascosto sia la collana che il braccialetto, oppure no?

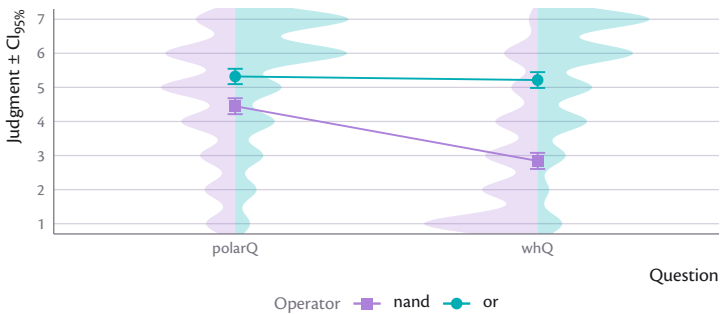
No, non ha nascosto sia la collana che il braccialetto, ma non ricordo quale non abbia nascosto.



(per nulla accettabile) 1 2 3 4 5 6 7 (totalmente accettabile)

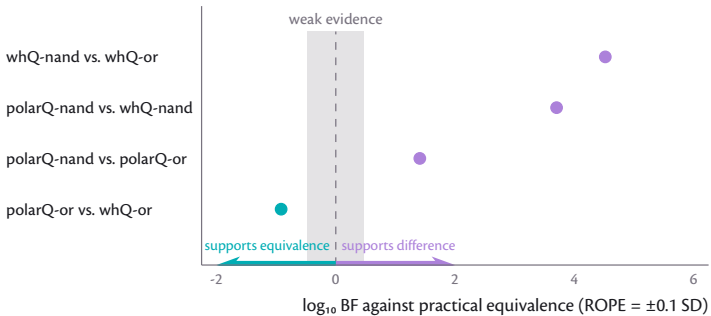
polarQ-nand condition

Results



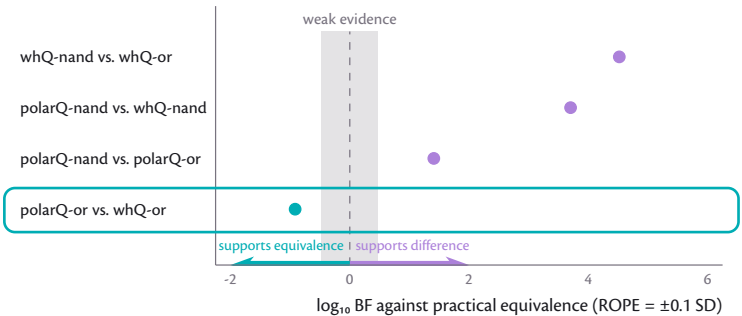
raw results; shaded areas are rating distributions

Bayes factors



cumulative probit model, maximal random effects for participants and items

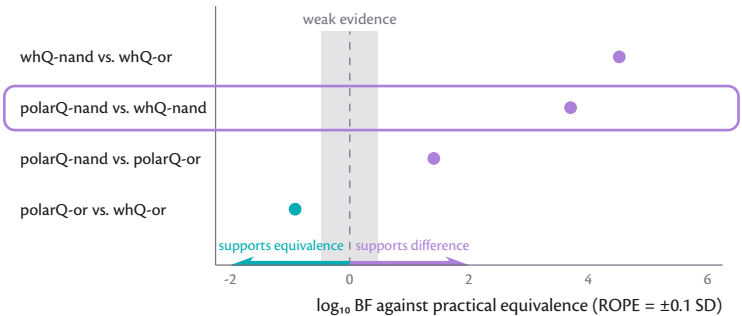
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or:
no difference

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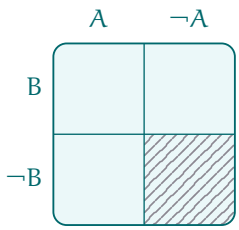
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nand:
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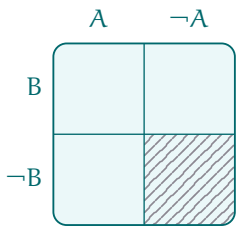
An asymmetry



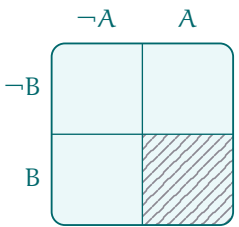
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answer: $A \vee B$

- *Which ones did he hide?* $A \vee B$ is relevant: it rules out one cell;

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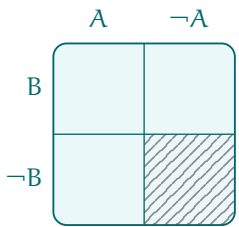
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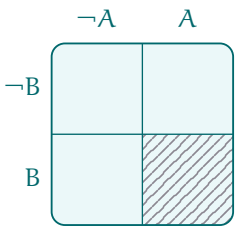
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- *Which ones did he hide?* $A \vee B$ is relevant: it rules out one cell;
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- *Which ones did he hide?* $A \vee B$ is relevant: it rules out one cell;
- *Which ones did he not hide?* $\neg(A \wedge B)$ is relevant in the same way;
- But negated conjunction is unable to answer a *wh*-question the way disjunction can.

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Two focus placements for negated conjunction

(6) He didn't have both [the apple juice]$_F$ and [the beer]$_F$.

(7) He [didn't]$_F$ have both the apple juice and the beer.

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- Focus on the conjuncts: only $\neg A \wedge \neg B$;
- Polarity focus: $\neg(A \wedge B)$ is fine.

The experiment shows that negation can scope over conjunction with polar questions.

Focus: a second dimension of meaning

Rooth (1992)

The meaning of a phrase containing focus is a pair: $\langle \text{ordinary value}, \{\text{alternatives}\} \rangle$.

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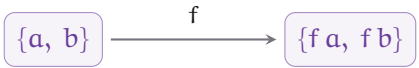
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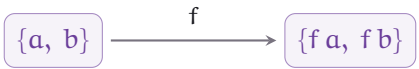


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- The alternatives at the root are the question the sentence answers.



Polarity focus

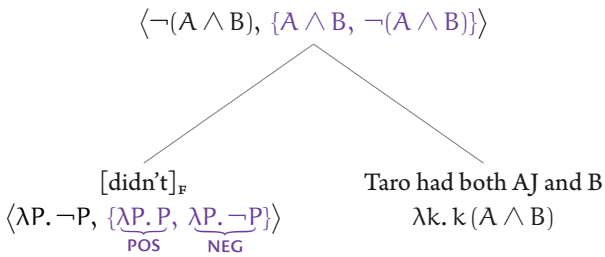
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Polarity focus

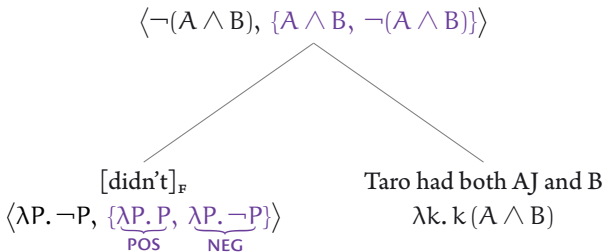
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Taro had both AJ and B
$$\lambda k.\, k\,(A \wedge B)$$

Polarity focus

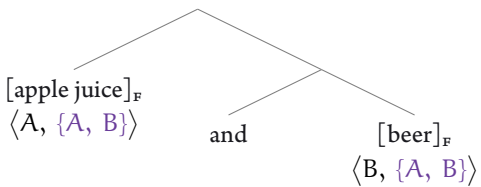


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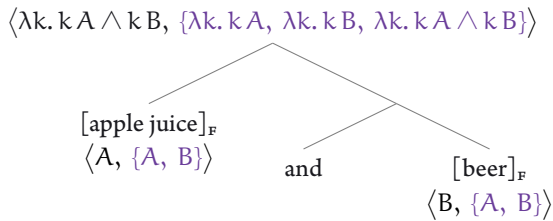


alternatives $\{A \wedge B,\ \neg(A \wedge B)\}$: a polar question

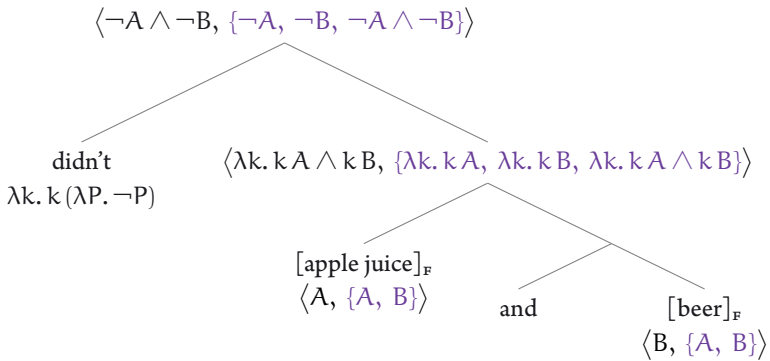
Focus on the conjuncts: wide scope



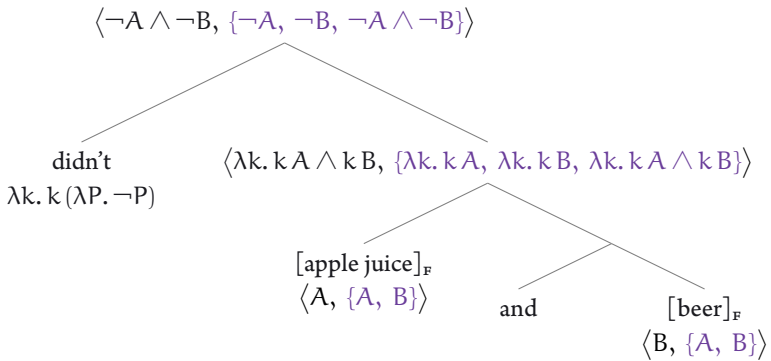
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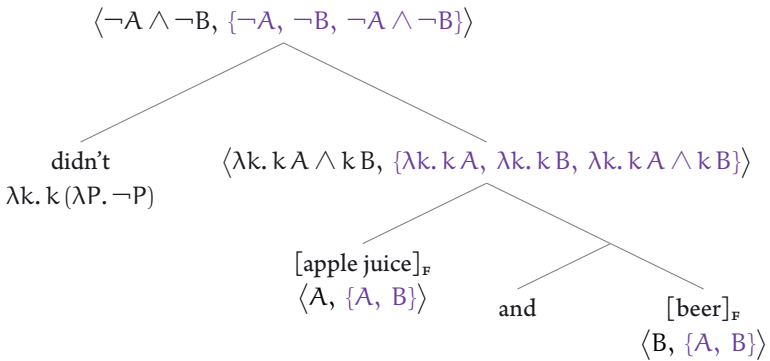


Focus on the conjuncts: wide scope



wide scope of conjunction

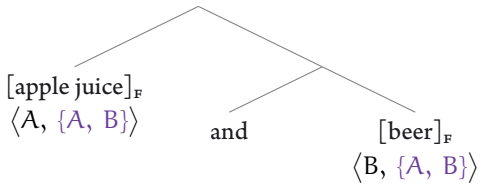
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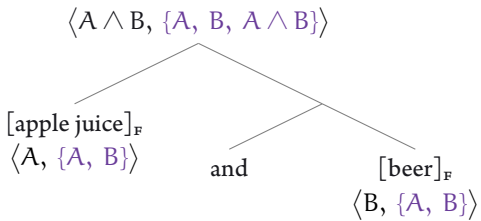
wide scope of conjunction

The alternatives at the root are the *wh*-question.

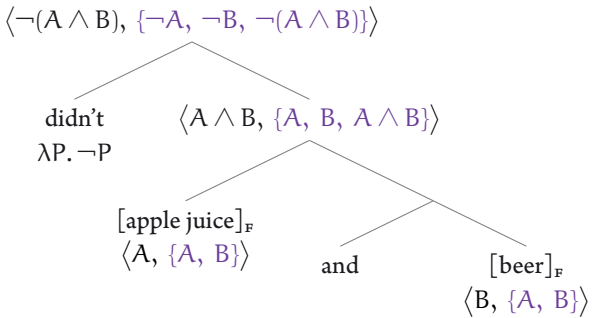
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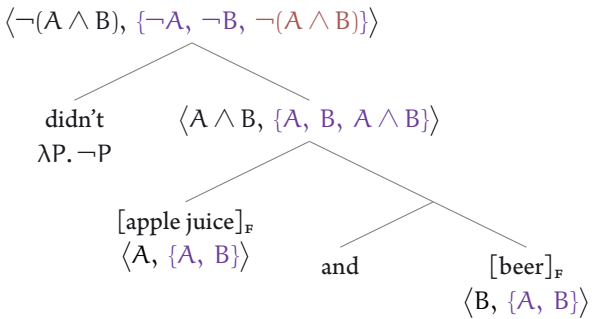
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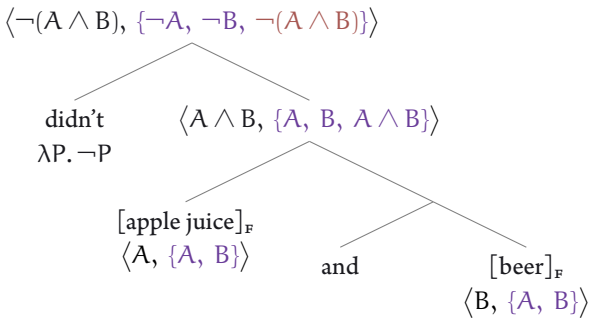


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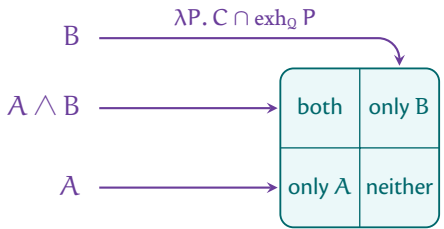
Is this still an answer to the *wh*-question?

Question-partition matching

QPM: A question is well-formed only if exhaustifying each alternative in the context C picks out a cell of its own, and every cell is picked out this way. *Fox 2018*

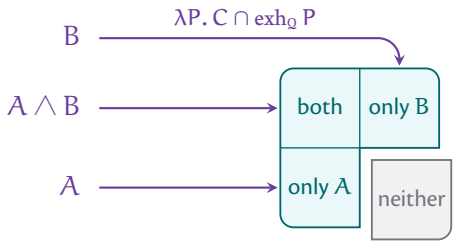
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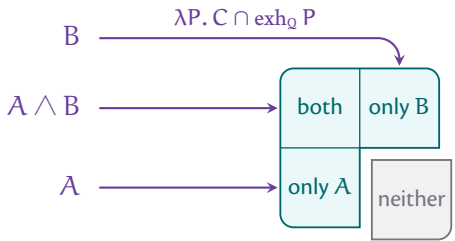


Did Taro have the apple juice, the beer, or both?

It has to be presupposed that he had one of the two.

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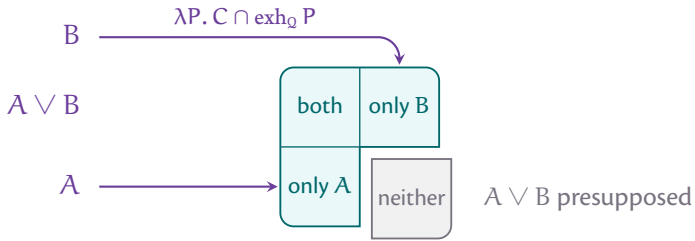
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Focus alternatives have to match a question, so QPM constrains them too.

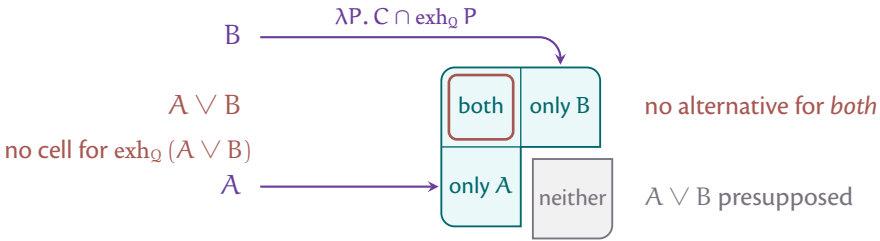
A case where matching fails

Adding disjunctions of alternatives to the set makes QPM fail.



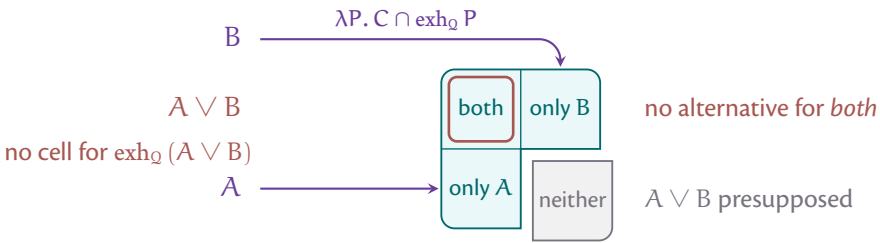
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This can make negated conjunction fail QPM:

$$\{\neg A, \neg B, \neg(A \wedge B)\} = \{\neg A, \neg B, \neg A \vee \neg B\}$$

Four derivations

	scope and focus	alternatives	QPM
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- $\neg(A \wedge B)$ only via (d): an answer to the polar question;
- (c) does not match any question that passes QPM.

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Each is a good answer to the same question; $A \vee B$ is never an alternative.

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$$\neg\Diamond A$$

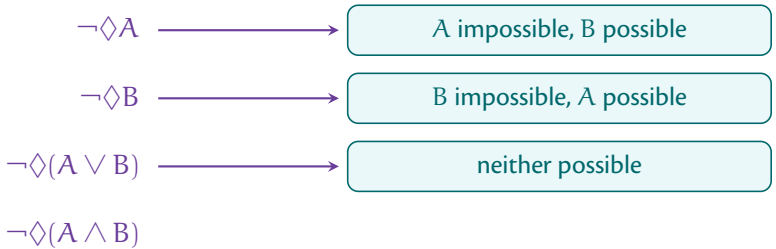
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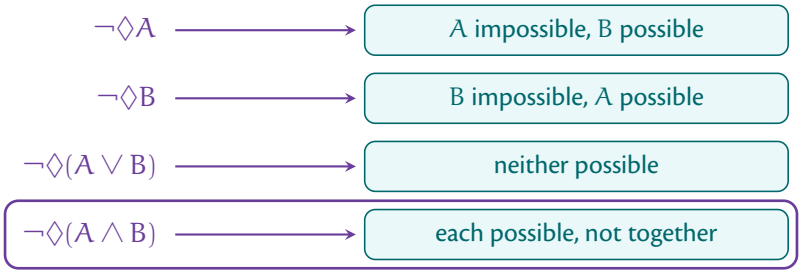
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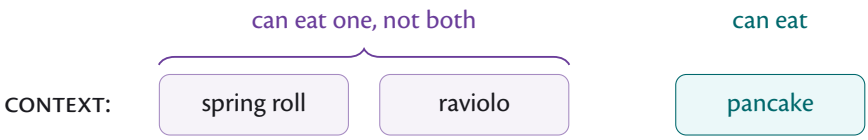


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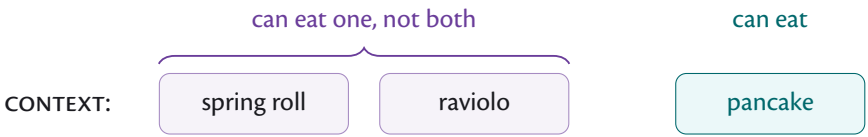
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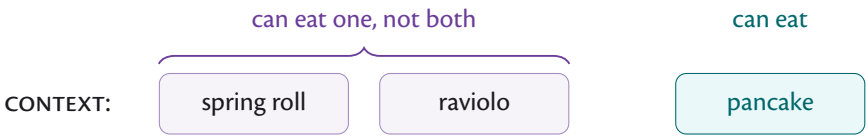


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(8) Q: Which leftovers is Peter not allowed to eat?
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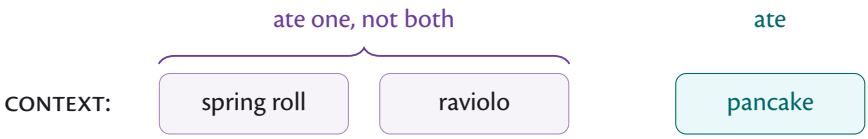


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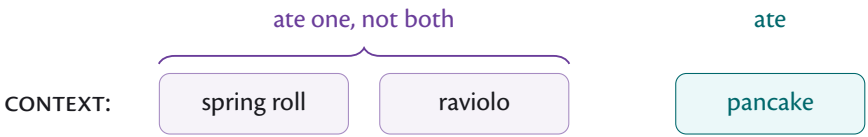
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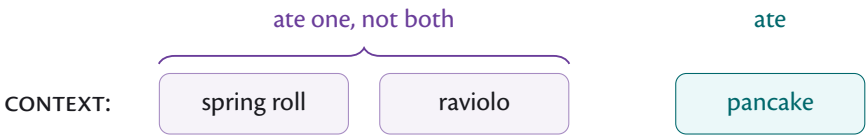
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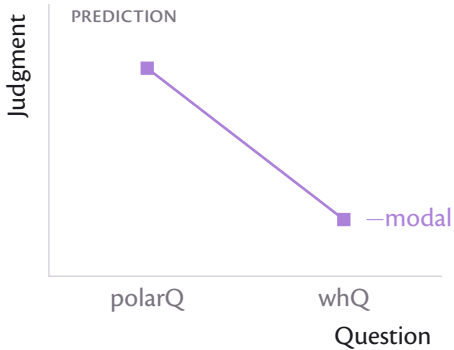
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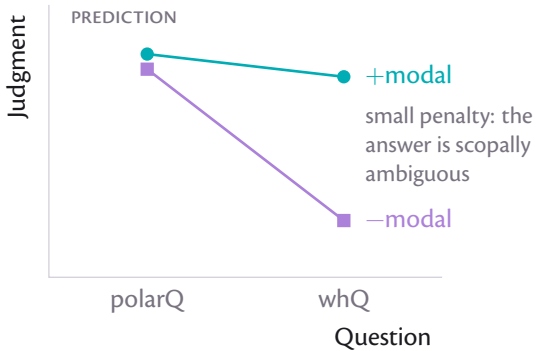
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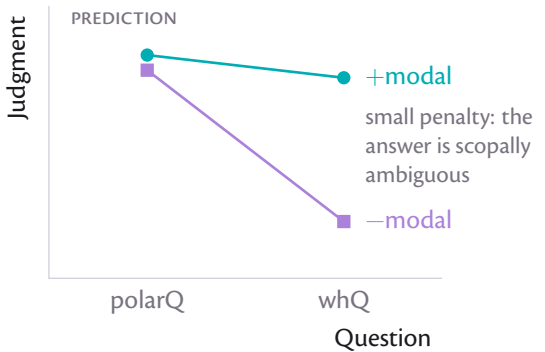
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The predicted contrast is hard to derive without structured answers.

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to be tested soon in project B04

References

Alonso-Ovalle, Luis (2006). "Disjunction in alternative semantics". PhD thesis. Amherst, MA: UMass Amherst.

Benbaji-Elhadad, Ido & Omri Doron (2026). "The Varieties of Relevance: Over-informative answers and the case for keeping relevance strong". In: *Journal of Semantics*. Accepted (with revisions).

Ciardelli, Ivano & Floris Roelofsen (2015). "Alternatives in Montague Grammar". In: *Proceedings of Sinn und Bedeutung 19*. Ed. by Eva Csipak & Hedde Zeijlstra, pp. 161–178.

Feinmann, Diego (2026). "A note on relevance". In: *Journal of Semantics* 43.1, pp. 1–15. DOI: 10.1093/jos/ffaf013.

Fox, Danny (2018). "Partition by exhaustification: Comments on Dayal 1996". In: *Proceedings of Sinn und Bedeutung 22*. Ed. by Uli Sauerland & Stephanie Solt. Vol. 1. ZASPiL 60. Berlin, pp. 403–434. DOI: 10.21248/zaspil.60.2018.474.

Groenendijk, Jeroen & Martin Stokhof (1984). "Studies on the semantics of questions and the pragmatics of answers". PhD thesis. Amsterdam: University of Amsterdam.

Lewis, David (1988). "Statements partly about observation". In: *Philosophical Papers* 17.1, pp. 1–31. DOI: 10.1080/05568648809506282.

Roberts, Craige (2012). "Information structure in discourse: Towards an integrated formal theory of pragmatics". In: *Semantics and Pragmatics* 5, pp. 1–69. DOI: 10.3765/sp.5.6.

Rooth, Mats (1992). "A theory of focus interpretation". In: *Natural Language Semantics* 1.1, pp. 75–116. DOI: 10.1007/bf02342617.

Model coefficients

